Data Science and Analytics

T.Y.B.Sc. (C.S.) — Semester V Practical Examination

Complete Answers: Q1, Both Q2 Options, and Viva

Q1. DataFrame and Scatter Plot (10 Marks)

Aim: Create a DataFrame containing car models, engine sizes, and fuel efficiency, then visualize the relationship with a scatter plot.

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Car Model': ['Swift', 'Honda City', 'Creta', 'Baleno', 'Fortuner'],
    'Engine Size': [1.2, 1.5, 1.6, 1.2, 2.8],
    'Fuel Efficiency': [22, 18, 17, 23, 12]
}

df = pd.DataFrame(data)
print(df)

plt.scatter(df['Engine Size'], df['Fuel Efficiency'], color='red')
plt.title('Engine Size vs Fuel Efficiency')
plt.xlabel('Engine Size (Litres)')
plt.ylabel('Fuel Efficiency (km/L)')
plt.show()
```

Result: The DataFrame is created and the scatter plot shows the relationship between engine size and fuel efficiency.

Note: These are example values. Use the data supplied by your college if it is provided.

Q2 — Option 1. Simple Linear Regression (20 Marks)

Aim: Predict Performance Score from Training Hours Per Week and evaluate the model using the R^2 score.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = {
    'Training Hours': [2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
    'Performance Score': [40, 45, 50, 55, 60, 65, 70, 75, 80, 85]
}

df = pd.DataFrame(data)
print(df)

X = df[['Training Hours']]
y = df['Performance Score']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("R2 Score:", r2_score(y_test, y_pred))

plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, model.predict(X), color='red', label='Regression Line')
plt.title('Training Hours vs Performance Score')
plt.xlabel('Training Hours Per Week')
plt.ylabel('Performance Score')
plt.legend()
plt.show()
```

Result: The model is trained, the R^2 score is displayed, and a regression line is plotted.

Note: The dataset values above are sample values because the actual college dataset was not provided.

Q2 — Option 2. KNN Classifier (20 Marks)

Aim: Classify customers into segments using K-Nearest Neighbors and evaluate the model using accuracy and a confusion matrix.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

data = {
    'Age': [20, 25, 30, 35, 40, 45, 22, 28, 38, 48],
    'Spending Score': [80, 70, 60, 50, 40, 30, 85, 65, 45, 25],
    'Segment': ['A', 'A', 'B', 'B', 'C', 'C', 'A', 'B', 'C', 'C']
}

df = pd.DataFrame(data)
print(df)

X = df[['Age', 'Spending Score']]
y = df['Segment']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

new_customer = pd.DataFrame([[32, 55]],
                            columns=['Age', 'Spending Score'])
new_customer = scaler.transform(new_customer)
print("New Customer Segment:", model.predict(new_customer)[0])
```

Result: The KNN classifier predicts customer segments and displays accuracy and the confusion matrix.

Note: This is sample customer data. Replace it with the dataset supplied by your college.

Q3. Viva Questions and Answers (5 Marks)

1. What is Data Science? The process of collecting, analyzing, and interpreting data to obtain useful information.

2. What is a DataFrame? A two-dimensional data structure with rows and columns in Pandas.

3. What is Pandas? A Python library used for data manipulation and analysis.

4. What is Matplotlib? A Python library used to create graphs and charts.

5. What is a scatter plot? A graph that shows the relationship between two variables using points.

6. What is Machine Learning? A branch of AI that enables computers to learn patterns from data.

7. What is Linear Regression? A supervised learning algorithm used to predict continuous values.

8. What is the R^2 score? A measure of how well a regression model explains variation in the target variable.

9. What is KNN? K-Nearest Neighbors; it predicts a class using nearby data points.

10. What is classification? A machine-learning task that predicts categories or classes.

11. What is accuracy? The proportion of correct predictions among all predictions.

12. What is a confusion matrix? A table comparing actual classes with predicted classes.

13. What is preprocessing? Preparing raw data for analysis or machine learning.

14. What is StandardScaler? A tool that standardizes numeric features to mean about zero and standard deviation about one.

15. What is train_test_split()? A function that divides data into training and testing sets.